

UI_REDESIGN_ROADMAP.md

VSSC Infrastructure Management Portal

Frontend Redesign Roadmap & Implementation Playbook

Version: 1.0

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PURPOSE OF THIS DOCUMENT

This document is the implementation blueprint for the complete frontend redesign of the VSSC Infrastructure Management Portal.

Unlike the other documentation files which explain the existing system, this document explains:

- What the final system should become
- Why the redesign is being performed
- What stages exist
- What has already been completed
- What remains to be built
- How future implementation should proceed

This document acts as the master execution roadmap.

REDESIGN OBJECTIVE

Transform the application from:

Traditional CRUD Dashboard

into:

Infrastructure Operations Center

without modifying:

- Database schema
- Existing backend APIs
- Existing synchronization workflows
- Existing authentication mechanisms

This project is a frontend transformation, not a backend rewrite.

DESIGN PHILOSOPHY

The application should feel similar to:

- Infrastructure Operations Consoles
- Enterprise Monitoring Platforms
- Internal Engineering Tools
- Datacenter Management Systems

The application should NOT resemble:

- Generic Admin Templates
 - Consumer Web Applications
 - Marketing Websites
 - AI Chat Applications
-

USER EXPERIENCE GOALS

The redesigned application should allow users to answer:

Infrastructure Questions

- How many VMs exist?
 - How many nodes exist?
 - Which clusters are active?
 - What storage resources are available?
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Ownership Questions

- Who owns a VM?
- Which division consumes the most resources?
- Which entity has the highest allocation?

- Which users are attached to a VM?

Capacity Questions

- How much RAM is allocated?
- How much CPU is allocated?
- How much storage is allocated?
- What trends are emerging?

Monitoring Questions

- Which nodes are under pressure?
- Which VMs are consuming resources?
- What infrastructure requires attention?

TARGET INFORMATION ARCHITECTURE

Dashboard

Infrastructure

- └ Clusters
- └ Nodes
- └ Storage

Virtual Machines

- └ Inventory
- └ Details

Monitoring

- └ Pressure Analytics
- └ Resource Trends

Analytics

- └ Entity Usage
- └ Division Usage
- └ Capacity Projection

Reports

- └ Report Builder
- └ Report History

Administration

- └─ User Management
- └─ Sync Center
- └─ VM Requests

Settings

- └─ Profile
- └─ Theme
- └─ Password

DESIGN SYSTEM

Theme Support

Required:

- Light Theme
- Dark Theme

Already Implemented.

Typography

Primary:

Noto Serif

Fallback:

Liberation Serif

Final Fallback:

Times New Roman

Reason:

- RHEL compatibility
 - Professional appearance
 - Dashboard readability
-

Color System

Light

Background

#F8FAFC

Surface

#FFFFFF

Primary

#1D4ED8

Accent

#2563EB

Dark

Background

#0B1220

Surface

#111827

Primary

#2563EB

Accent

#3B82F6

COMPONENT STANDARDS

Cards

- Rounded corners
- 16px–18px radius
- Soft shadows

Tables

- Sticky headers
- Hover states
- Responsive layouts

Forms

- Validation
- Accessibility
- Theme awareness

Icons

- Lucide React
-

MOTION SYSTEM

Use:

Framer Motion

Allowed Animations:

- Fade
- Slide
- Scale
- Stagger

Required:

- Page transitions

- Card transitions
- Sidebar transitions
- Drawer transitions
- Modal transitions

Animations should enhance usability, not distract from it.

ROLE EXPERIENCE MODEL

ADMIN EXPERIENCE

Purpose:

Infrastructure Operator

Capabilities:

- Full infrastructure visibility
 - Synchronization
 - User management
 - VM ownership management
 - Reporting
 - Analytics
 - Monitoring
-

MANAGER EXPERIENCE

Purpose:

Operational Observer

Capabilities:

- View infrastructure
- View VMs
- View monitoring
- View analytics
- Generate reports

Restrictions:

- No synchronization
- No user creation
- No ownership modification

USER EXPERIENCE

Purpose:

Resource Consumer

Capabilities:

- View assigned VMs
- View personal analytics
- View personal reports

Restrictions:

- No global visibility
- No infrastructure access
- No administrative access

Data visibility is derived through:

vm_user_relation

COMPLETED STAGES

STAGE 0

System Reverse Engineering

Status:

Completed

Deliverables:

- Repository analysis
 - Workflow analysis
 - Database analysis
 - API analysis
-

STAGE 1

Foundation Refactor

Status:

Completed

Deliverables:

- Application shell
 - Sidebar
 - Topbar
 - Theme provider
 - Design tokens
 - Route architecture
 - Motion foundation
-

STAGE 1.1

Infrastructure Integration

Status:

Completed

Deliverables:

- Node integration
 - Cluster integration
 - Storage integration
 - Real backend connectivity
-

STAGE 1.2

Analytics Foundation

Status:

Completed

Deliverables:

- Entity analytics
 - Division analytics
 - Trend analysis
 - Capacity projections
-

STAGE 1.3

RBAC Foundation

Status:

Completed

Deliverables:

- Protected routes
 - Role-aware navigation
 - User filtering
-

STAGE 1.4

Role Validation & Sync Verification

Status:

Current

Objectives:

- Verify sync workflows
- Verify role restrictions

- Verify ownership filtering
- Verify route protection

Acceptance Criteria:

- Users only see assigned VMs
 - Managers have read-only global access
 - Sync operations function correctly
 - Protected routes work correctly
-

REMAINING STAGES

STAGE 2

Dashboard Refinement & Infrastructure Center

Priority:

High

Purpose:

Create a true operational dashboard.

Dashboard

Implement:

KPI Cards

- Total VMs
 - Running VMs
 - Stopped VMs
 - Total Nodes
 - Total Clusters
 - Total Storages
-

Visualizations

- VM Status Distribution

- VM Distribution By Cluster
 - VM Distribution By Node
-

Ownership Analytics

- Entity Distribution
 - Division Distribution
 - Group Distribution
-

Nodes Screen

Implement:

- Health indicators
 - Memory visualization
 - CPU visualization
 - Status cards
 - Node table redesign
-

Clusters Screen

Implement:

- Cluster summary cards
 - Cluster distribution charts
 - Infrastructure statistics
-

Storage Screen

Implement:

- Capacity visualizations
 - Storage status indicators
 - Storage analytics
-

Acceptance Criteria:

Infrastructure screens feel production-ready.

STAGE 3

VM Management Redesign

Priority:

Highest

Purpose:

Transform VM inventory into the primary operational interface.

VM Inventory

Features:

- Advanced search
 - Multi-column filtering
 - Sorting
 - Status indicators
-

VM Details Drawer

Tabs:

Overview

VM specifications

Ownership

Assigned personnel

Storage

Storage mappings

Technical

Advanced metadata

Acceptance Criteria:

VM workflows become significantly easier than the original implementation.

STAGE 4

Monitoring Center

Purpose:

Operational monitoring.

Pressure Analytics

CPU Pressure

Memory Pressure

IO Pressure

Resource Trends

VM growth

CPU growth

RAM growth

Storage growth

Acceptance Criteria:

Monitoring data is easily understandable and actionable.

STAGE 5

Advanced Analytics

Purpose:

Infrastructure intelligence.

Entity Usage

- VM Count
 - CPU Allocation
 - RAM Allocation
 - Storage Allocation
-

Division Usage

- Resource allocation
 - Comparative analysis
-

Capacity Projection

Trend-based estimates only.

No AI forecasting.

No machine learning claims.

Label projections as:

Estimated Capacity Trend

Acceptance Criteria:

Analytics are derived entirely from existing schema.

STAGE 6

Reports & Administration

Purpose:

Administrative modernization.

Reports

Report Builder

Report History

Exports

Sync Center

Improved synchronization UX.

Show:

- Progress
 - Success
 - Errors
-

User Management

Modern administration interface.

Acceptance Criteria:

Administrative workflows become transparent and user-friendly.

STAGE 7

User Experience Refinement

Purpose:

Dedicated user-facing experience.

My Dashboard

Assigned VM overview.

My Virtual Machines

Filtered inventory.

My Analytics

Personal resource usage.

My Reports

Personal exports.

Acceptance Criteria:

Users never see infrastructure unrelated to them.

STAGE 8

Final Polish

Purpose:

Production readiness.

Tasks:

Accessibility

Performance

Loading states

Skeleton loaders

Error states

Cross-browser testing

RHEL validation

Animation refinement

Responsive validation

Acceptance Criteria:

Application feels production-grade.

IMPLEMENTATION RULES

Every future stage must:

1. Respect database constraints.
 2. Respect API contracts.
 3. Preserve existing workflows.
 4. Maintain role boundaries.
 5. Use reusable components.
 6. Follow the design system.
 7. Support light and dark themes.
 8. Maintain accessibility.
 9. Remain compatible with RHEL deployment.
 10. Be reviewed before moving to the next stage.
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ANTIGRAVITY WORKFLOW

For every future implementation:

Step 1

Read:

AI_PROJECT_CONTEXT.md

Step 2

Read:

SYSTEM_REVERSE_ENGINEERING.md

Step 3

Read:

DATABASE_SCHEMA_ANALYSIS.md

Step 4

Read:

API_REFERENCE.md

Step 5

Read:

UI_REDESIGN_ROADMAP.md

Step 6

Determine current stage.

Step 7

Implement only the current stage.

Never skip stages.

Step 8

Provide:

- Files modified
 - Files created
 - Architectural decisions
 - Screenshots
 - Known limitations
-

FINAL SUCCESS DEFINITION

The redesign is successful when:

- Infrastructure visibility is significantly improved.
- Navigation is intuitive.
- Analytics are meaningful.
- Monitoring is actionable.
- User roles are properly separated.
- Visual design is modern and professional.
- Existing backend functionality remains unchanged.
- The platform feels like a dedicated Infrastructure Operations Center suitable for operational use within VSSC.